# Project a Reusable Spatial MEM Basis

## Description

Projects exact or fast training MEMs to held-out locations with the matching construction method.

## Usage

```
project_spatial_mev_basis(
  list_spatial_mev_basis = NULL,
  data_coords_projected_pred = NULL,
  data_coords_projected_train = NULL,
  spatial_scale_attributes = NULL,
  projection_chunk_size = 5000L,
  exact_projection_function = interpolate_mev_to_grid,
  fast_projection_function = NULL
)
```

## Arguments

| | |
|---|---|
| `list_spatial_mev_basis` | List returned by [compute_spatial_mev_basis()]. |
| `data_coords_projected_pred` | Projected coordinates for prediction locations. |
| `data_coords_projected_train` | Training coordinates required for exact IDW compatibility. |
| `spatial_scale_attributes` | Optional training scale attributes passed to [scale_predictors_with_training_attributes()]. |
| `projection_chunk_size` | Positive integer limiting prediction rows processed in one operation. |
| `exact_projection_function` | Exact-basis projection function. Defaults to [interpolate_mev_to_grid()]. |
| `fast_projection_function` | Fast-basis projection function. Defaults to [spmoran::meigen0()]. |

## Value

Data frame of projected 'mev_*' columns in prediction-row order.